

OPTION B INSTITUTE FAMILY

M01 - DATA FOUNDATIONS + ESSENTIAL EXCEL

Standalone Zero-to-Job-Ready Data Engineer 2026

BEGINNER-FIRST MASTER REPAIR RC4

15 canonical lessons | full 16-stage teacher contract | protected attempt-first
review

This learner book is rebuilt from the frozen canonical technical material. It is not a front-matter patch: every lesson is re-laid out into the shared institute teaching grammar while preserving the course competency, source route, guided work, independent proof and validation boundary.

This book is your teacher

Use the book in order. External videos and official docs are visual companions, not hidden prerequisites. Watch or read only the assigned scope, then close it before the independent task.

The 16-stage lesson loop

Why it matters -> What you learn -> Prerequisite -> Picture it -> New words -> Useful source -> Follow once -> Expected result -> Why it worked -> Beginner mistakes -> Recovery -> Independent task -> Validate -> Teach it back -> Protected review + changed retry -> Evidence/mastery + exact next route.

Attempt-first protection

Never open a protected lesson review before saving a genuine independent attempt. If repair is needed, compare only the weak idea, close the review, solve a changed retry, then validate again. A solution seen during review is repair evidence, not mastery.

What counts as mastery

A reproducible artifact or explanation, an independent validation control, preserved failure/recovery evidence, a changed transfer task completed without the worked answer open, and an own-words explanation. Percent-complete is not a substitute.

How to use your devices

Phone: assigned video/reference, reading, revision. Laptop/PC: every real command, query, file, workbook, code task, validation and evidence artifact. Install only what the current lesson asks for.

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Prerequisite / tool map

Understand data grain/quality and use essential Excel without damaging source data.

Item	Rule
Entry prerequisite	Previous module/Gate evidence must be stable.
Primary device	PC/laptop for hands-on work; phone is support only.
External sources	Optional/supporting unless the lesson explicitly marks an official current reference.
Evidence	Module practical + fresh retest + saved evidence; no critical unresolved failure.

Practice / review / Gate route

Lesson teaching -> follow once -> independent attempt -> validate -> save evidence. Only after the genuine attempt may protected review open. If weak: targeted repair -> close review -> changed retry -> revalidate. Finish the module practical/retest route before any downstream Gate that depends on this module.

DE01-01 - What data is: tables, records, fields and data types

1. Why it matters

Data tools cannot be used safely until you know what one row and each field mean.

2. What you are learning

Understand what a table, row, column, cell and data type are before you touch formulas or SQL.

3. Prerequisite / starting state

Open Raw_Sales. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Think of a school attendance register. Each student has one row.

5. New words before we use them

Word Plain-English meaning Table A rectangular collection of related data. Row / record One observation or event at the table's current level of detail. Column / field One attribute stored consistently across rows. Cell One value at the intersection of a row and column. Data type The kind of value: text, whole number, decimal, date/time, true/false, etc.

6. Useful video / source

Source: Luke Barousse — Data Engineer Bootcamp (2026)

Assigned scope: 4:16:18 Data Modeling & JOINS + 6:57:49 Data Types

How to use it: Use the mapped chapters as visual reinforcement for tables/fields/types. The Teaching Book supplies the beginner definitions and independent proof.

Note: Current 2026 long-form DE bootcamp; exact chapter starts are published in the video description.

[Open source](#)

7. Follow with me once

- Open the workbook sheet Raw_Sales.
- Look only at the header row first: OrderID, OrderDate, CustomerID, CustomerName, Region, Product, Qty,
- UnitPrice, PaymentMethod.
- Pick row 2. Read it left to right as one order record.
- Classify Qty and UnitPrice as numbers, OrderDate as a date-like value, and Region/Product/PaymentMethod as text
- categories.
- G01 - Map table grain and field types

8. Expected result / what you should see

You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- A value looking like a date does not guarantee Excel stored it as a true date.
- Formatting is appearance; data type is how the value behaves.

- Do not assume every number should be added. OrderID contains digits but is an identifier, not a quantity.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- A value looking like a date does not guarantee Excel stored it as a true date.
- Formatting is appearance; data type is how the value behaves.
- Do not assume every number should be added. OrderID contains digits but is an identifier, not a quantity.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P01 - Grain and candidate key In Raw_Sales, choose OrderID, Qty, UnitPrice and PaymentMethod. Write whether each is identifier, numeric measure or text category, and explain why.

13. Validate

You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 1 fresh grain/key prompt Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 1 fresh grain/key prompt succeeds without the worked lesson/review open. Exact next route: DE01-02

DE01-02 - Grain: what one row represents

1. Why it matters

Grain controls counting, duplicates, joins and summaries; wrong grain creates believable but wrong answers.

2. What you are learning

Learn grain - the sentence that says exactly what one row represents.

3. Prerequisite / starting state

Open Raw_Sales + Support_Tickets. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A photo can show one person, one family or one whole crowd. A table also has a level of detail.

5. New words before we use them

Word Plain-English meaning Grain The business meaning of one row. Order header Usually one row per order. Order line Usually one row per product line inside an order. Aggregation Combining many lower-level rows into fewer summary rows.

6. Useful video / source

Source: Teaching RC2 visual teaching + hands-on lab

How to use it: Use the Teaching Book and guided lab. No generic video is forced here; mastery still requires the independent task.

Note: This lesson is better learned from the Teaching Book + hands-on demonstration than from a generic external video.

Disposition: COURSE_BUILT / BOOK_ONLY for this lesson; no filler link is required.

7. Follow with me once

- Look at Raw_Sales. Ask: if OrderID O1001 appears once, what event does that row describe?
- Write: One row represents one order record for one customer/product purchase in this practice data.
- Now imagine an order-line table where one order buys three products. That same OrderID could legitimately appear
- three times because the grain is one order line, not one order.
- G01 - Map table grain and field types

8. Expected result / what you should see

Before counting, joining or summarizing, you can say "one row represents ..." in plain English.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Saying "grain = customer" just because CustomerID exists.
- Counting rows as orders without proving the table is one row per order.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Saying "grain = customer" just because CustomerID exists.
- Counting rows as orders without proving the table is one row per order.

- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P01 - Grain and candidate key + P02 - Legitimate repeats versus duplicates For Support_Tickets, write the grain sentence. Then explain why CustomerID may repeat even if TicketID should not.

13. Validate

Before counting, joining or summarizing, you can say "one row represents ..." in plain English.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Before counting, joining or summarizing, you can say "one row represents ..." in plain English.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open.

Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 1 fresh grain/key prompt Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 1 fresh grain/key prompt succeeds without the worked lesson/review open. Exact next route: DE01-03

DE01-03 - Keys, duplicates, NULLs and missing data

1. Why it matters

A repeated value is only a duplicate relative to the table grain and key. Missingness also needs a business rule, not automatic deletion.

2. What you are learning

Tell legitimate repeats from suspicious duplicates and understand missing values.

3. Prerequisite / starting state

Open Raw_Sales + Support_Tickets. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A person can buy five times, so their customer ID can repeat. But a single receipt number printed twice for the exact same sale is suspicious.

5. New words before we use them

Word Plain-English meaning Key A column or combination of columns used to identify the row at its intended grain. Duplicate A repeated row/key that should be unique at that grain. NULL / blank A missing or unknown value. Zero A known numeric value equal to nothing - not the same as missing.

6. Useful video / source

Source: Luke Barousse — Data Engineer Bootcamp (2026)

Assigned scope: 4:16:18 Data Modeling & JOINS + 10:16:15 Date/Text/NULL Functions

How to use it: Watch only the relevant chapter blocks. Teaching Book adds duplicate detection and the course validation task.

Note: Support source, not a replacement for the course duplicate/NULL exercise.

[Open source](#)

7. Follow with me once

- In Raw_Sales, find the two O1019 rows. Compare every column. They are identical, so this is suspicious at
 - one-order-record grain.
- Find CustomerID C003 appearing on more than one order. That is not automatically wrong because one customer
 - can place multiple orders.
- Find the blank CustomerID and blank PaymentMethod. Treat these as missing-information questions, not as zero.
- G02 - Key and duplicate inspection + G03 - Missingness and validity scan

8. Expected result / what you should see

You do not delete a repeat until you can explain why it violates the key/grain.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Using "Remove Duplicates" before deciding the key.
- Replacing blanks with 0 just to make them disappear.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
 - continue.

11. If you get stuck

- Using "Remove Duplicates" before deciding the key.
- Replacing blanks with 0 just to make them disappear.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P02 - Legitimate repeats versus duplicates In Raw_Sales, write: candidate key, one legitimate repeat, one suspicious duplicate, and two missing-value issues.

13. Validate

You do not delete a repeat until you can explain why it violates the key/grain. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You do not delete a repeat until you can explain why it violates the key/grain.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 1 duplicate/missing reasoning Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 1 duplicate/missing reasoning succeeds without the worked lesson/review open. Exact next route: DE01-04

DE01-04 - Data quality dimensions and validation thinking

1. Why it matters

Quality checks turn vague trust into explicit tests that can fail visibly before bad data reaches reporting.

2. What you are learning

Turn "data quality" from a vague phrase into specific checks.

3. Prerequisite / starting state

Open Raw_Sales. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

Before accepting a food delivery you might check: did everything arrive, is anything broken, is it the right item, did you receive it on time? Data quality works the same way.

5. New words before we use them

Word Plain-English meaning Completeness Are required values present? Validity Do values obey allowed rules/ranges? Uniqueness Are values that should be unique actually unique? Consistency Do the same concepts use consistent labels/formats? Timeliness Is the data recent enough for the decision?

6. Useful video / source

Source: Statistics Canada — Data Quality in Six Dimensions

Assigned scope: Full 8:42 beginner video

How to use it: Watch the short official video, then use the Teaching Book to apply quality dimensions to engineering-style validation.

Note: Official government video; beginner-friendly and stable conceptual material.

[Open source](#)

7. Follow with me once

- Completeness: check CustomerID/PaymentMethod blanks.
- Validity: Qty should normally be positive; Region should belong to the approved set.
- Uniqueness: OrderID should be unique if grain is one order.
- Consistency: East and east should not silently become different business regions.
- Timeliness: an old dataset may be correct but too stale for today's operational decision.
- G03 - Missingness and validity scan

8. Expected result / what you should see

Every quality claim becomes a concrete test: column + rule + what failure looks like.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Writing "check data quality" without a test.
- Confusing uniqueness with "no repeated values anywhere".
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Writing "check data quality" without a test.
- Confusing uniqueness with "no repeated values anywhere".
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P03 - Design five data-quality checks Using Raw_Sales, write one specific check for each of the five dimensions.

13. Validate

Every quality claim becomes a concrete test: column + rule + what failure looks like. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Every quality claim becomes a concrete test: column + rule + what failure looks like.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 1 concrete quality-test design Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 1 concrete quality-test design succeeds without the worked lesson/review open. Exact next route: DE01-05

DE01-05 - Meet Excel: workbook, worksheet, cells and formulas

1. Why it matters

Excel instructions depend on precise workbook, worksheet and cell language; navigation mistakes later become formula mistakes.

2. What you are learning

Understand the Excel screen: workbook, worksheet, cell, address, formula bar and formula.

3. Prerequisite / starting state

Open START_HERE + workbook working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A workbook is like a notebook file. Worksheets are the tabs/pages inside it.

5. New words before we use them

Word Plain-English meaning Workbook The .xlsx file. Worksheet A tab inside the workbook. Cell address Column letter + row number, such as D2. Formula bar The area that shows the actual value/formula stored in the active cell. Formula An instruction beginning with = that Excel calculates.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 1:15–4:09

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Open M01_EXCEL_PRACTICE_WORKBOOK_TEACHING_RC2.xlsx.
- Click the Formula_Practice tab.
- Click cell D2. Notice the Name Box/selected cell address says D2.
- Type =B2*C2 and press Enter.
- Click D2 again. The cell shows a result, while the formula bar shows =B2*C2.
- G04 - Workbook navigation proof

8. Expected result / what you should see

You understand the difference between what the cell displays and the formula stored behind it.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Typing 1700 manually instead of =B2*C2.
- Typing B2*C2 without the leading =.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Typing 1700 manually instead of =B2*C2.

- Typing B2*C2 without the leading =.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

Checkpoint 2 - navigation/formula setup proof Click three different cells and say their addresses aloud. Then write one simple addition formula in an unused cell and undo it.

13. Validate

You understand the difference between what the cell displays and the formula stored behind it.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You understand the difference between what the cell displays and the formula stored behind it.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 2 closed-book workbook navigation Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 2 closed-book workbook navigation succeeds without the worked lesson/review open. Exact next route: DE01-06

DE01-06 - Enter, save and structure tabular data safely

1. Why it matters

Preserving the source and using a working copy makes mistakes reversible and keeps evidence reproducible.

2. What you are learning

Enter and save tabular data without destroying the original source.

3. Prerequisite / starting state

Open Raw_Sales working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A raw file is like an original photograph. Edit a copy so you can always return to the untouched original.

5. New words before we use them

Word Plain-English meaning Raw data Original received data before your transformations. Working copy A separate file you are allowed to edit. Header row The row containing column names. Tabular structure One header row, one observation per row, one attribute per column.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 4:09–8:21

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Open the workbook. Use File > Save As and create a working copy such as M01_working.xlsx.
- Keep one header row. Avoid merged cells inside data tables.
- Do not insert decorative blank rows in the middle of the dataset.
- Save frequently with Ctrl+S.
- G04 - Workbook navigation proof + safe working-copy discipline

8. Expected result / what you should see

The original workbook remains unchanged, and your working copy contains your experiments.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Editing the only copy of the raw file.
- Using two header rows or merged cells as decoration inside source data.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Editing the only copy of the raw file.
- Using two header rows or merged cells as decoration inside source data.

- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

Checkpoint 2 - safe working-copy and structure proof Create a working copy, rename it clearly, and write the full filename in your evidence notes.

13. Validate

The original workbook remains unchanged, and your working copy contains your experiments. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The original workbook remains unchanged, and your working copy contains your experiments.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 2 preserve source + working-copy proof Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 2 preserve source + working-copy proof succeeds without the worked lesson/review open. Exact next route: DE01-07

DE01-07 - Sort and filter without damaging data

1. Why it matters

Filtering should change only what you see; sorting should change order without breaking row integrity.

2. What you are learning

Use Sort and Filter as viewing/ordering tools without deleting or misaligning data.

3. Prerequisite / starting state

Open Raw_Sales working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A filter is like temporarily hiding books that do not match your search; the books are still on the shelf.

5. New words before we use them

Word Plain-English meaning Filter Temporarily shows rows matching conditions; hidden rows remain in the table. Sort Changes row order based on one or more columns. Whole-row integrity Values from the same record must move together.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 11:37–15:46

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- In a working copy of Raw_Sales, click inside the data.
- Choose Data > Filter.
- Filter Region to East/east and note the visible rows.
- Clear the filter and confirm the full set returns.
- For sorting, click inside the table and sort the entire data range by UnitPrice - do not select only UnitPrice cells in
 - isolation.
 - G05 - Safe sort/filter

8. Expected result / what you should see

Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Deleting hidden rows because they are not visible.
- Sorting one column by itself and scrambling row relationships.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
 - continue.

11. If you get stuck

- Deleting hidden rows because they are not visible.
- Sorting one column by itself and scrambling row relationships.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P04 - Filter and prove no deletion Filter one PaymentMethod, note the visible count, clear it, and confirm the original row count returns.

13. Validate

Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 2 filter/restore + safe sort Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 2 filter/restore + safe sort succeeds without the worked lesson/review open. Exact next route: DE01-08

DE01-08 - Basic formulas and essential functions

1. Why it matters

Formulas make calculations repeatable and auditable; manual arithmetic is used as an independent control, not as the production method.

2. What you are learning

Create basic formulas and use SUM, AVERAGE and COUNT/COUNTA.

3. Prerequisite / starting state

Open Formula_Practice. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A formula is a recipe. If the ingredients change, Excel should recalculate the result automatically.

5. New words before we use them

Word Plain-English meaning Cell reference A pointer to a cell such as B2. SUM Adds numeric values. AVERAGE Calculates arithmetic mean. COUNT Counts numeric cells. COUNTA Counts nonblank cells.

6. Useful video / source

Source: Chandoo — Excel Tutorial for Beginners

Assigned scope: 15:46–22:39

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Open Formula_Practice. In D2 type =B2*C2.
- Fill/copy D2 down through the product rows.
- Manually check Keyboard: $2 \times 850 = 1700$.
- Use =SUM(D2:D7) for total Sales.
- Use =AVERAGE(D2:D7) for average Sales.
- Use COUNT on a numeric column or COUNTA on a populated text/ID column depending on what you intend to
 - count.
- G06 - Basic row formulas

8. Expected result / what you should see

Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Typing answers instead of formulas.
- Using COUNT on text IDs and thinking zero means there are no records.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
 - continue.

11. If you get stuck

- Typing answers instead of formulas.
- Using COUNT on text IDs and thinking zero means there are no records.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P05 - Revenue and Margin formulas Complete Formula_Practice and manually validate two rows before trusting the total.

13. Validate

Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 2 fresh Sales formula + manual check Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 2 fresh Sales formula + manual check succeeds without the worked lesson/review open. Exact next route: DE01-09

DE01-09 - Relative and absolute references

1. Why it matters

Relative references should move with rows; constants such as a tax/rate cell must stay fixed when copied.

2. What you are learning

Understand why A2 moves when copied but \$H\$2 stays fixed.

3. Prerequisite / starting state

Open Reference_Practice. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A relative reference is like "the house next door" - it changes when you move. An absolute reference is like a full GPS coordinate - it stays fixed.

5. New words before we use them

Word Plain-English meaning Relative reference A2; changes relative to the formula location when copied. Absolute reference \$H\$2; row and column are locked. Mixed reference \$H2 or H\$2; only one part is locked.

6. Useful video / source

Source: Chandoo — Useful Excel concepts

Assigned scope: 12:35–20:52

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Open Reference_Practice. Calculate Base Sales in D2 using Qty x UnitPrice.
- Tax Rate is stored once in H2. In E2 use a formula such as $=D2*(1+\$H\$2)$.
- Copy E2 down. Click E3, E4, E5. D2 changes to D3/D4/D5, but \$H\$2 stays exactly \$H\$2.
- Temporarily change H2. Every taxed result should update because all formulas point to the same rate cell. Undo if
- needed.
- G07 - Relative vs absolute reference

8. Expected result / what you should see

One setting cell controls every row without repeating the percentage in each formula.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Using H2 without dollar signs and watching it drift to H3/H4.
- Typing 18% directly in every row formula.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Using H2 without dollar signs and watching it drift to H3/H4.

- Typing 18% directly in every row formula.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P06 - One absolute rate cell Complete Independent_Rate using the single rate in H2 and explain what each dollar sign protects.

13. Validate

One setting cell controls every row without repeating the percentage in each formula. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. One setting cell controls every row without repeating the percentage in each formula.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 3 fresh absolute-reference task Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 3 fresh absolute-reference task succeeds without the worked lesson/review open. Exact next route: DE01-10

DE01-10 - Excel Tables and structured data habits

1. Why it matters

Excel Tables create explicit rectangular structure, stable headers and predictable expansion/filter behavior.

2. What you are learning

Know the difference between a normal range and an actual Excel Table.

3. Prerequisite / starting state

Open clean working range. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A normal range is a group of cells. An Excel Table is a managed object that knows its headers, rows and filters.

5. New words before we use them

Word Plain-English meaning Excel Table A structured Excel object created with Insert > Table or Ctrl+T. Header Named field at the top of a Table column. Structured behavior Filters, automatic expansion and formula propagation tied to the Table object.

6. Useful video / source

Source: Chandoo — Useful Excel concepts

Assigned scope: 2:22–8:45

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Work on a clean copy/range. Click inside it.
- Choose Insert > Table (or Ctrl+T).
- Confirm the full range and tick My table has headers.
- Notice filter arrows appear. Add one new row directly below the Table and see the Table expand.
- G08 - Excel Table behaviour

8. Expected result / what you should see

The object expands with new rows and keeps explicit headers/filters.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Assuming coloured formatting means the data is an Excel Table.
- Creating a Table from an incomplete range that omits columns.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Assuming coloured formatting means the data is an Excel Table.
- Creating a Table from an incomplete range that omits columns.

- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P07 - Prove Excel Table behaviour Convert a small clean range to a Table, add a row, then describe one behavior that proves it is a Table.

13. Validate

The object expands with new rows and keeps explicit headers/filters. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. The object expands with new rows and keeps explicit headers/filters.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 3 fresh Table behavior proof Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 3 fresh Table behavior proof succeeds without the worked lesson/review open. Exact next route: DE01-11

DE01-11 - Practical cleaning and lookup awareness

1. Why it matters

Cleaning should standardize representation without inventing business meaning; lookups connect tables by a deliberate key.

2. What you are learning

Understand basic cleaning and how XLOOKUP connects related tables using an ID.

3. Prerequisite / starting state

Open Raw_Sales / PowerQuery_Source + Customer_Lookup. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A lookup is like using a roll number to find a student's house from a separate register. The ID is the matching key.

5. New words before we use them

Word Plain-English meaning Cleaning Making values consistent/usable without changing their business meaning. Lookup value The ID you are trying to find. Lookup array Where Excel searches for that ID. Return array The column from which Excel returns the answer. Not found The explicit result when no match exists.

6. Useful video / source

Source: Chandoo — XLOOKUP for Beginners

Assigned scope: Whole short video

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Notice Region values East, east and West with trailing space in Raw_Sales. These are consistency problems.
- Open Customer_Lookup. CustomerID is the linking field; Tier is the value you want to bring back.
- In Independent_Lookup, use XLOOKUP with CustomerID as the lookup value, Customer_Lookup!CustomerID as search range, and Tier as return range.
- Handle C999 explicitly with a not-found message rather than silently assuming a tier.
- G09 - Cleaning awareness + lookup follow-along in learner book

8. Expected result / what you should see

Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Matching by CustomerName when CustomerID is the intended key.
- Treating unmatched IDs as blank and forgetting to investigate them.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Matching by CustomerName when CustomerID is the intended key.
- Treating unmatched IDs as blank and forgetting to investigate them.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P08 - Lookup Tier by CustomerID Complete Independent_Lookup, spot-check two known customers, and explain the four pieces of XLOOKUP in plain English.

13. Validate

Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 3 fresh lookup with missing-ID handling Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 3 fresh lookup with missing-ID handling succeeds without the worked lesson/review open. Exact next route: DE01-12

DE01-12 - PivotTable and Power Query awareness

1. Why it matters

PivotTables summarize quickly; Power Query records repeatable import/cleaning logic. At this stage you need a safe mental model, not advanced BI mastery.

2. What you are learning

Understand what PivotTables and Power Query are for without trying to master them yet.

3. Prerequisite / starting state

Open Pivot_Source + PowerQuery_Source. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

PivotTable is a fast summary machine. Power Query is a recorded cleaning/import recipe you can replay.

5. New words before we use them

Word Plain-English meaning PivotTable Interactive summary built from source data. Rows area Categories listed vertically. Values area Measures summarized, e.g., Sum of Sales. Power Query Excel's repeatable import/transform tool. Refresh Run the saved import/transform logic again on updated data.

6. Useful video / source

Source: Chandoo — PivotTable / Power Query awareness

Assigned scope: 31:35–38:35

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

- Pivot: click Pivot_Source > Insert > PivotTable > New Worksheet. Put Region in Rows and Sales in Values.
- Check that Values says Sum of Sales, not Count of Sales.
- Power Query awareness: inspect PowerQuery_Source. Find leading/trailing spaces, inconsistent case, mixed date
- formats and Amount stored as text with comma.
- The point is not to hand-fix every cell. A Power Query workflow can record trim/clean/type/standardization steps and
- refresh them later.
- Pivot follow-along in learner book + P09 after guided attempt

8. Expected result / what you should see

You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than "another worksheet".

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Building a fancy dashboard before understanding the summary.
- Using Count of Sales by accident.

- Manually cleaning source values one-by-one and calling that a repeatable pipeline.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Building a fancy dashboard before understanding the summary.
- Using Count of Sales by accident.
- Manually cleaning source values one-by-one and calling that a repeatable pipeline.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P09 - Pivot summary and reconciliation Create the simple PivotTable, reconcile its grand total to source SUM, then write 2-3 sentences explaining Power Query.

13. Validate

You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than "another worksheet". Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than "another worksheet".

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 4 fresh Pivot summary + source reconciliation Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 4 fresh Pivot summary + source reconciliation succeeds without the worked lesson/review open. Exact next route: DE01-13

DE01-13 - Chart choice and basic visual communication

1. Why it matters

A chart should answer the business question clearly; decorative visuals can hide the comparison instead of explaining it.

2. What you are learning

Choose a simple chart that matches the question and avoid decorative charts that hide the message.

3. Prerequisite / starting state

Open Pivot_Source / summary range. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A chart is a sentence in picture form. Choose the picture that makes the comparison easiest to read.

5. New words before we use them

Word Plain-English meaning Bar/column chart Good for comparing categories. Line chart Good for trends over ordered time. Pie chart Limited use; difficult when many categories/slices exist. Axis Scale along horizontal/vertical direction. Title Should state what is being compared, not just "Chart 1".

6. Useful video / source

Source: Chandoo — What is a Chart and How to Make Charts in Excel (video)

Assigned scope: Embedded beginner video: chart purpose, create/format/change chart type; use the page's "select correct chart" follow-up only as needed

How to use it: Watch only enough to understand chart choice and basic communication. Then use the Teaching Book task with your own data; do not turn this into a dashboard course.

Note: Older page, but chart-selection fundamentals are stable. Final link audit must recheck that the embedded video still plays.

[Open source](#)

7. Follow with me once

- Use the regional summary from your PivotTable.
- For comparing East/West/North/South, choose a bar or column chart.
- Give it a clear title such as "Sales by Region".
- Check that the largest bar visually matches the largest numeric total.
- Simple chart follow-along + Checkpoint 4 visual task

8. Expected result / what you should see

A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Using a line chart for unordered categories.
- Adding 3-D effects that distort comparisons.
- Making the title vague.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then

- continue.

11. If you get stuck

- Using a line chart for unordered categories.
- Adding 3-D effects that distort comparisons.
- Making the title vague.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

Checkpoint 4 - create a simple comparison chart Create one simple regional comparison chart and write one sentence explaining why that chart type fits the question.

13. Validate

A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 4 fresh chart-choice task Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 4 fresh chart-choice task succeeds without the worked lesson/review open. Exact next route: DE01-14

DE01-14 - Reconciliation, sanity checks and source preservation

1. Why it matters

Plausible output is not proof. Reconciliation and source preservation make a result independently checkable and recoverable.

2. What you are learning

Learn to distrust even plausible-looking results until you reconcile them to an independent check.

3. Prerequisite / starting state

Open completed working copy + source data. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

If a shop cashier says the total is 3810, you may still count the item totals separately. Reconciliation is that second route to the answer.

5. New words before we use them

Word Plain-English meaning Reconciliation Comparing the result to an independent total/count/control. Source preservation Keeping raw evidence unchanged. Sanity check A quick reasonableness check for impossible or surprising values. Control total A known count/amount used to verify processing.

6. Useful video / source

Source: Teaching RC2 visual teaching + hands-on lab

How to use it: Use the Teaching Book and guided lab. No generic video is forced here; mastery still requires the independent task.

Note: This lesson is better learned from the Teaching Book + hands-on demonstration than from a generic external video.

Disposition: COURSE_BUILT / BOOK_ONLY for this lesson; no filler link is required.

7. Follow with me once

- Before transformations, record source row count and any important total.
- After filtering/cleaning/calculating, recompute the total another way.
- If Pivot grand total differs from SUM of source Sales, stop and find the earliest step where rows/values diverged.
- Do not "fix the final number" manually; repair the logic.
- Reconciliation follow-along + Checkpoint 4

8. Expected result / what you should see

You can show both the result and one independent check that supports it.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Trusting a pretty PivotTable because it looks professional.
- Changing raw data to force agreement.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

11. If you get stuck

- Trusting a pretty PivotTable because it looks professional.
- Changing raw data to force agreement.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then
- continue.

12. Small independent task

P09 + Checkpoint 4 - reconcile to independent total For Pivot_Source, compare Pivot grand total with =SUM(E2:E19). Record both and explain whether they reconcile.

13. Validate

You can show both the result and one independent check that supports it. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. You can show both the result and one independent check that supports it.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Checkpoint 4 independent reconciliation + limitation Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 4 independent reconciliation + limitation succeeds without the worked lesson/review open. Exact next route: DE01-15

DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain -> save evidence

1. Why it matters

A professional workflow is a chain: inspect, calculate, validate, explain and save evidence. Skipping any link makes handoff fragile.

2. What you are learning

Combine the module into a repeatable beginner workflow: inspect -> calculate -> validate -> explain -> save evidence.

3. Prerequisite / starting state

Open one completed M01 working file. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

4. Picture it in your head

A good analyst is not someone who can press many Excel buttons. It is someone who can explain what data they received, what they changed, what result they produced, and why they trust it.

5. New words before we use them

Word Plain-English meaning Inspect Understand columns, grain, keys and quality before calculating. Calculate Apply formulas/summaries that answer the question. Validate Use an independent check. Explain State result and reasoning in plain language. Evidence Saved workbook, note, screenshot, formula or output proving what you actually did.

6. Useful video / source

Source: Teaching RC2 visual teaching + hands-on lab

How to use it: Use the Teaching Book and guided lab. No generic video is forced here; mastery still requires the independent task.

Note: This lesson is better learned from the Teaching Book + hands-on demonstration than from a generic external video.

Disposition: COURSE_BUILT / BOOK_ONLY for this lesson; no filler link is required.

7. Follow with me once

- Read the business question first.
- State the table grain and candidate key.
- Preserve raw source and work on a copy.
- Perform only the transformations/calculations needed.
- Validate with row counts, control totals or spot checks.
- Explain the answer, one limitation and one possible failure mode.
- Save the workbook and note where the evidence lives.
- Module workflow rehearsal before practical

8. Expected result / what you should see

Someone else can reproduce and audit your work instead of trusting an unexplained final number.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Starting with a chart/formula before understanding the data.
- Saving only screenshots and losing the actual workbook/formulas.

- Reporting "done" with no validation.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

11. If you get stuck

- Starting with a chart/formula before understanding the data.
- Saving only screenshots and losing the actual workbook/formulas.
- Reporting "done" with no validation.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

12. Small independent task

P10 - Mini analyst handoff + Module Practical Write a 5-8 sentence handoff for one M01 practice task: source, grain, what you changed, result, validation, limitation.

13. Validate

Someone else can reproduce and audit your work instead of trusting an unexplained final number. Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task. Someone else can reproduce and audit your work instead of trusting an unexplained final number.

14. Teach it back

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Module Practical; if weak, different-domain Fresh Retest Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Module Practical; if weak, different-domain Fresh Retest succeeds without the worked lesson/review open. Exact next route: Proceed to the module practical/Gate route after the module mastery checklist.

Module mastery checklist + revision quick reference

- ☐ Every lesson has a saved genuine independent attempt.
- ☐ Every lesson has at least one independent validation/control appropriate to the tool.
- ☐ Protected reviews were opened only after attempts; any weak lesson has a changed retry after review was closed.
- ☐ You can teach back each lesson in your own words and name one realistic failure mode.
- ☐ Required module practical/retest evidence is saved and no critical defect is unresolved.
- ☐ D+1, D+4, D+10 and D+30 revision events are scheduled for weak or high-value competencies.

Revision cadence

D+1: fresh micro-task. D+4: mixed retrieval. D+10: changed/failure case. D+30: interview/project retrieval. Always attempt from memory before reopening a source.

Release boundary

This book is a static learner artifact. It does not prove live tool/cloud execution or a protected Gate pass. Runtime evidence must be recorded separately when the curriculum requires it.